

09-10-3

2.(1){c3,d3}

(2){a1,b1,c1,d1},{a1,b1,c2,d2},{a2,b2,c1,d1},{a2,b2,c2,d2},
{a3,b3,c1,d1},{a3,b3,c2,d2}

(3) $R \div S = \{a1,b1\}, \{a2,b2\}, \{a3,b3\}$

$R \div S = \{m | m \in \{t[12] | t \in r\} \text{ AND } \neg (m \in \{t[12] | t \in \{ \{k | k \in \{h[1234]\} | h[12] \in \{t[12] | t \in r\} \text{ AND } h[34] \in s\} \text{ AND } \neg (k \in r)\} \})\}$

3.

(1) SELECT sname FROM Sailors, Boats, Reserves WHERE

Sailors.sid = Reserves.sid AND Boats.bid = Reserves.bid

AND Reserves.bid > 103 AND Boats.color = '蓝'

(2) SELECT bname FROM Boats WHERE color = '蓝' AND bid IN

(SELECT bid FROM Reserves GROUP BY bid HAVING COUNT(DISTINCT sid) = 1)

(3) SELECT sid FROM Reserves

WHERE bid IN (SELECT bid FROM Boats WHERE color = '蓝')

GROUP BY sid

WHERE COUNT(DISTINCT bid) = (SELECT COUNT(*) FROM Boats WHERE color = '蓝')

(4) SELECT Y.master FROM Sailor, Reserves

WHERE Sailor.sid = Reserves.sid

GROUP BY master

Having COUNT(DISTINCT bid) >= all(

SELECT count(DISTINCT bid) FROM Sailor, Reserves

WHERE Sailor.sid = Reserves.sid

GROUP BY master

)

7.

调度 1 17 错误调度 $R2(A,x)$ 与 $W1(A,x-1)$ 为冲突操作，不能交换执行顺序

调度 2 19 错误调度 $R2(A,x), W2(A,x-3)$ 与 $W1(A,x-1)$ 都为冲突操作，不能交换执行顺序

调度 3 16 正确调度

只有调度为可串行化调度时才正确，即冲突操作的顺序不允许更改

8.

CREATE TRIGGER insert_on_sailors

BEFORE INSERT ON Sailors

REFERENCING NEW AS N

FOR EACH ROW

WHEN (EXISTS(SELECT * FROM Sailors GROUP BY master HAVING COUNT(*) >= 2))

ROLLBACK;

附加题:

(1) sid+bid

(2) 职工号+零件号, 仓库号+零件号

10-11

4.

(1) SELECT 姓名, 仓库位置 FROM 仓库管理员, 仓库, 采购订单
WHERE 仓库管理员.库管员编号 = 采购订单. 库管员编号
仓库.库管员编号 = 仓库管理员.库管员编号
AND 服装编号 = '0101'

(2) SELECT 服装编号 FROM 采购订单 GROUP BY 服装编号
HAVING COUNT(DISTINCT 库管员编号) = 1

(3) SELECT 姓名, 库管员编号 FROM 仓库管理员 WHERE 库管员编号 IN
(SELECT 库管员编号 FROM 采购订单 GROUP BY 库管员编号 HAVING
COUNT(DISTINCT 供应商编号) = SELECT COUNT(*) FROM 供应商)

(4)

SELECT Y.库管员编号, Y. 订货日期, Y.应到货日期 FROM 采购订单 Y,
(SELECT 库管员编号, MAX(a) b, c FROM (SELECT 库管员编号, SUM(服装数量) a,
MAX(服装数量) c 采购订单 FROM GROUP BY 库管员编号)) T
WHERE Y.库管员编号 = T.库管员编号 AND Y.服装数量 = T.c

14-15-3

3.(1) 客人 关系模式中 主键为身份证号

房间 关系模式中 主键为房间号

住宿 关系模式中 主键为 (身份证号, 房间号, 入住时间)

(2) A. SELECT 客人.身份证号 FROM 客人, 住宿

WHERE 客人.身份证号 = 住宿. 身份证号

AND 籍贯 = '南京' AND 房间号 = '301'

B. SELECT 身份证号 FROM 住宿 GROUP BY 身份证号

HAVING COUNT(DISTINCT 房间号) = (SELECT COUNT(*) FROM 房间)

C. SELECT 姓名, T.b FROM 客人 X,

(SELECT 身份证号, COUNT(房间号) b FROM 住宿 WHERE YEAR(入住时间) =
'2014' GROUP BY 身份证号

HAVING COUNT(房间号) >=

all(SELECT COUNT(*) FROM 住宿 WHERE YEAR(入住时间) = '2014'
GROUP BY 身份证号))T

WHERE X.身份证号 = T.身份证号

4.

```
1.SELECT SNAME FROM STUDENT WHERE SNO NOT IN
  (SELECT SNO FROM SC WHERE CNO = 'CS110')
2.SELECT SNAME FROM STUDENT WHERE SEX = '女' AND SNO IN
  (SELECT SNO FROM SC WHERE CNO LIKE 'EE%')
3.SELECT SNO, COUNT(*), MAX(GRADE), MIN(GRADE), AVG(GRADE) FROM SC GROUP
  BY SNO
4.SELECT SNAME FROM STUDENT WHERE SNO IN
  ( SELECT SNO FROM SC WHERE GRADE > 90 AND CNO IN
    (SELECT CNO FROM COURSE WHERE SEMESTER = '秋')
    GROUP BY SNO HAVING COUNT(*) > 2 )
5.SELECT SNAME,CNO FROM STUDENT,SC WHERE STUDENT.SNO = SC.SNO AND NOT
  EXISTS
  (SELECT * FROM SC,(SELECT CNO, AVG(GRADE) AS AVGGRADE FROM SC GROUP BY
    CNO) AS X
    WHERE STUDENT.SNO = SC.SNO AND SC.CNO = X.CNO AND SC.GRADE <
    X.AVGGRADE)
```

15-16-3

2.course 表的主键 cno+dname, 外键

Student 表的主键 sid,

Enroll 表的主键 sid+cno+dname1, 外键 sid,cno+dname1

3.

```
(1) SELECT sid, sname FROM Student WHERE sid IN
  (SELECT sid FROM Enroll WHERE grade > 3 AND dname1 = 'Computer Sci')
(2) SELECT cno,dname1 FROM Enroll GROUP BY cno, dname1 HAVING COUNT(sid) = 1
(3) SELECT cno, AVG(grade) FROM Enroll
  WHERE dname1 = 'Computer Sci' GROUP BY cno
(4) SELECT sname, sid FROM Student WHERE sid IN
  (SELECT sid FROM Enroll WHERE dname1 = 'Computer Sci' GROUP BY sid
  HAVING COUNT(*) =
  (SELECT COUNT(*) FROM course WHERE dname = 'Computer Sci'))
(5)
SELECT a, sid FROM Enroll, (
  SELECT cno, MAX(grade) a FROM Enroll WHERE dname1 = 'Computer Sci'
  GROUP BY cno) T
WHERE Enroll.grade = a AND Enroll.cno = T.cno AND dname1 = 'Computer Sci'
```

8.Create trigger insert_on_enroll

After Insert On enroll

Referencing NEW As N

For Each Statement

Insert Into failedcourse

Select sid, grade, cno, dname1, sectno, pname, dname2

From N

Where N.grade < 3.0